

MDFformer: A Multi-Scale Dual-Modal Fusion Transformer for Semantic Segmentation

Leixiong Shi, Yifei Dong, Tianqi Wang, Rui Cao and Xin Wen

Abstract—Remote sensing image interpretation is a crucial process for Earth observation and geoscience research, and multi-modal fusion techniques are key to improving semantic segmentation accuracy. CNN-based methods, due to the limited receptive field, suffer from inadequate long-range contextual modeling. While Transformer-based architectures alleviate the limitation, existing multi-modal fusion Transformers still face challenges in efficiently integrating multi-scale information. Therefore, this work proposes a multi-scale dual-modal fusion Transformer, namely MDFformer, to integrate RGB and DSM cross-modal interaction information. More specifically, MDFformer is a dual-stream encoder architecture combined with a multi-scale fusion FPN decoder. MDFformer introduces two novel modules: CSFM (Cross-Modality Shuffle Fusion Module) and SDEM (Spatial Detail Enhance Module), along with deep supervision (DS) to alter the gradient flow and enhance multi-scale feature learning. As a result, MDFformer efficiently utilizes both shallow and deep features from diverse modalities across multiple scales. The validation is performed on two publicly available high-resolution datasets, Vaihingen and Potsdam, achieving an overall accuracy (OA) of 92.06% and 91.25%, respectively. Therefore, the model effectively handles RGB and DSM data, making it suitable for remote sensing tasks and efficient earth sensing.

Index Terms—Digital surface model (DSM), Multi-scale, Multi-modality, Remote sensing, Semantic segmentation, Transformer

I. INTRODUCTION

With the advancement of geoscience research, precise and diverse Earth observation data are increasingly valued in fields such as urban planning [1] and land cover classification [2, 3]. Driven by the rapidly developing satellite sensors, which provide multisource data, and the limitations of single-modality models in handling complex surface features, multimodal fusion for remote sensing has become essential.

In this paper, we focus primarily on the semantic labeling task using both RGB and DSM data. RGB images possess excellent optical properties, while DSMs capture elevation information of all surface features. These distinct physical and numerical characteristics complement each other. However, existing models for multi-modal data fusion still face challenges, including difficulty in feature alignment and inefficient use of multi-scale information. In [4], two independent encoders first

generate modality-specific features for RGB and DSM separately. These features are then upsampled using a residual-like depth-wise separable convolution, allowing for effective extraction in high-resolution imagery. A similar approach is presented by H.A.H. Al-Najjar et al [5].

However, such CNN-based fusion methods struggle with long-range dependencies and global context modeling due to the limited receptive field inherent in convolutional operations. The success of transformer architecture alleviates the aforementioned difficulty, thanks to the self-attention mechanism. The STFF-GA [6] leverages attention to combine spatial-temporal information for remote sensing change detection. Similarly, the Vman model [7] introduces a visual-modified attention network to strengthen multimodal feature fusion, showing the flexibility and power of attention mechanisms across different data types. Emerging approaches such as Hyperrectangle Embedding for 3D scene prediction [8] and HE-3DSGR [9] further demonstrate the capability of Transformer-based models to tackle complex, structured data.

Moreover, single-modality models (e.g., RGB-only) lack elevation information critical for distinguishing surface features like buildings and low vegetation. CMFNet [10] is the first transformer-based network proposed for cross-modal fusion, which leverages a cross-modal attention mechanism to extract and fuse features. However, it struggles to handle fine-grained spatial representations effectively. To address this, FtransUNet [11] introduces adaptively mutually boosted attention, which incrementally boosts mutual information between modalities. Additionally, in CMX [12], spatial and channel weights are assigned separately to reduce noise interference.

However, those existing Transformer-based fusion models prioritize cross-modal interaction mechanisms but often overlook or inefficiently handle the critical aspect of multi-scale information integration. This limitation hampers their ability to accurately segment objects of vastly different sizes prevalent in remote sensing imagery.

In summary, This work proposes a multi-scale dual-modal fusion model, namely MDFformer. Through the dual-split cascade fusion process, the proposed CSFM achieves efficient cross-modal information fusion. Specifically, the module also incorporates the weight connections and channel shuffle to perform lightweight operations. The SDEM further enhances

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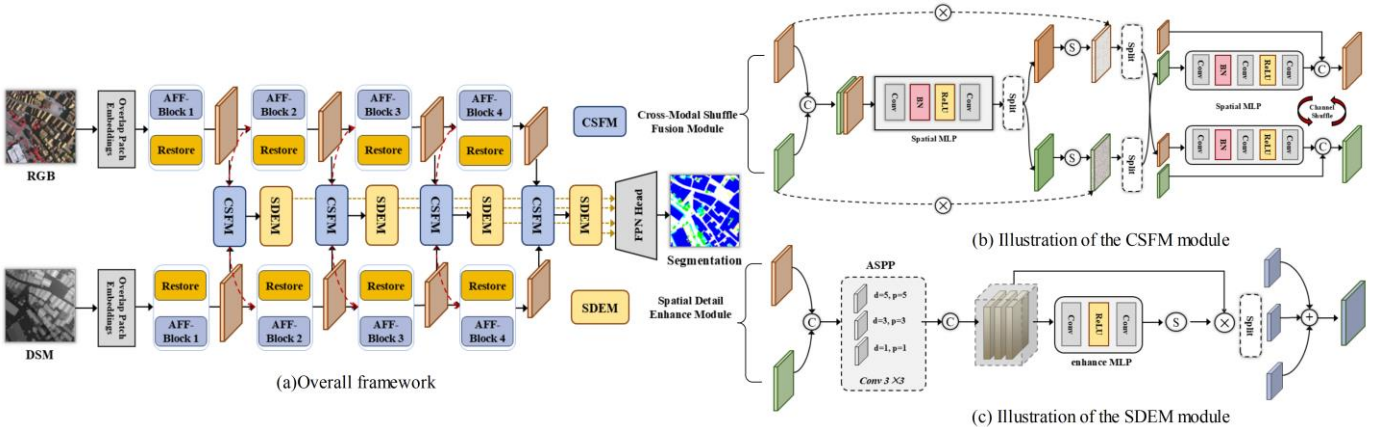


Fig. 1 An overview of the proposed MDFformer. It mainly consists of two components: CSFM (Cross-Modality Shuffle Fusion Module) and SDEM (Spatial Detail Enhancement Module).

spatial details by spatially mixing features extracted from multiple parallel atrous convolutions. The two novel modules used for intermediate processing receive deep supervision signals, which not only improve training efficiency but also enhance the model's ability to learn multi-scale features. Through this process, features from different levels of the backbone are sequentially shuffled, exchanged in CSFM, and enhanced in SDEM, enabling better interaction between different modal data. As a result, this architecture can effectively integrate detailed spatial information from the DSM and texture information from the RGB images, leveraging both shallow and deep features more efficiently. This letter makes the following contributions:

- (1) MDFformer demonstrates superior prediction performance on two benchmark datasets, achieving pixel-wise accuracies of 92.06% and 91.25% on the Vaihingen and Potsdam datasets, respectively.
- (2) The model exhibits low computational complexity, with 47.33G FLOPs and 35.82M parameters—outperforming most methods in efficiency and parameter economy.
- (3) This efficiency-accuracy trade-off confirms that our fusion strategy offers a lightweight and high-performance solution for remote sensing semantic segmentation.

II. PROPOSED METHOD

A. Overall Framework

MDFformer uses the AFFormer-base encoder [13], pre-trained on the ImageNet-1k, as its backbone. As shown in Fig. 1 (a), the process begins by embedding those into overlapping patch embeddings. These embeddings are then passed through a dual-stream encoder separately (four AFF-blocks, each processing both RGB and DSM features in parallel).

The core of the fusion process is handled by two key modules: the CSFM and SDEM. After each block, the features are sent through a branch into the CSFM to shuffle and interact with multi-scale features. The output of CSFM not only returns to the main branch but also passes through an SDEM to further enhance features. Finally, the multi-scale outputs from multiple SDEM modules are fed into an FPN head decoder.

Notably, the different scale feature maps fused by FPN during forward inference will generate their own additional loss.

During backpropagation, the gradient flows are influenced by both the total loss and the individual loss from each scale.

B. CSFM

The key to enhancing dense labeling accuracy lies in leveraging the complementary information. To achieve this, a novel module for cross-modal shuffling and fusion has been developed. The process begins by extracting features and concatenating them, and is followed by a series of operations that promote cross-modal interaction, as shown in Fig. 1 (b). Initially, R (RGB) and D (DSM) undergo channel concatenation. A Spatial MLP ϕ_1 is then applied to extract cross-modal interactions, helping to refine the feature representation. The channel split (CSP) divides the concatenated feature map into two parts, allowing each part to undergo independent weight calculation, yielding R_{weight} and D_{weight} . These weights are then multiplied by the original input to obtain R_{att} and D_{att} , where different parts exhibit distinct importance. This attention mechanism, while simple and lightweight, enables the model to achieve strong competitiveness through multi-scale attention acquisition, as demonstrated in B. Performance Comparison. Those operations can be represented as:

$$\begin{aligned}
 R_{weight}, D_{weight} &= \sigma \left(\text{CSP} \left(\phi_1([R, D]) \right) \right) \\
 R_{att} &= R_{weight} \odot R \\
 D_{att} &= D_{weight} \odot D
 \end{aligned} \tag{1}$$

Where $\sigma(\cdot)$ is the sigmoid function, $\odot$ represents element-wise multiplication, and $[\cdot]$ is the channel concatenation.

The next steps are divided into parallel operations, here taking the RGB as an example: the second split operation, the spatial split (SSP), divides the spatial dimensions of R_{att} into different regions, allowing for more localized interactions across the spatial domain:

$$R_{up}, R_{down} = \text{SSP}(R_{att}) \tag{2}$$

Similarly, D_{up} and D_{down} after SSP can be obtained. After this, R_{down} and D_{up} are exchanged. The exchanged then pass through another spatial MLP function, denoted as ϕ_2 , to extract higher-level spatial interactions. Finally, the feature maps are concatenated spatially and undergoes channel shuffle (CSH), ensuring that information flows between different groups of channels. The output after those operation is represented as:

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$$R_{out} = \text{CSH}(\langle \phi_2(D_{up}), R_{up} \rangle) \quad (3)$$

Where $\langle \cdot \rangle$ is spatial concatenation. In conclusion, through a concatenation-split-concatenation process, CSFM achieves cross-modal information fusion by integrating channel-level and spatial-level interactions.

C. SDEM

The SDEM is used to enhance the spatial details of the feature maps that pass through the CSFM module, ensuring better spatial representation. As shown in Fig. 1 (c), the process also starts by concatenating R-D and then using a series of convolutional and attention mechanisms to enhance the spatial features. Multi-scale processing is conventionally implemented using a pyramid structure, a strategy we also adopt. The compressed feature maps are processed through multiple parallel atrous convolutions (ASPP). These features are then concatenated, followed by compression through an enhance MLP, denoted as ϕ_3 . A sigmoid activation function is applied to compute the attention weights, ultimately resulting in the attention-weighted feature map. Those operations are given by:

$$\begin{aligned} T_c &= \text{ASPP}([R, D]) \\ T_e &= \sigma(\phi_3(T_c)) \odot T_c \end{aligned} \quad (4)$$

Where $\sigma(\cdot)$ $\odot$ and $[\cdot]$ perform the same function as in (1). The multi-scale enhanced texture T_e is split into three parts along the channel dimension. These parts correspond to the three dilated convolution outputs. After splitting, the parts are summed to combine the multi-scale texture features:

$$\begin{aligned} T_e &= [t_e^1, t_e^2, t_e^3] \\ out &= t_e^1 + t_e^2 + t_e^3 \end{aligned} \quad (5)$$

Where t_e^1, t_e^2, t_e^3 denote the feature maps split from T_e along the channel dimension, corresponding to the outputs of the three parallel atrous convolutions.

D. Deep Supervision

To further enhance the fusion at different levels, a method similar to depth supervision has been added on top of this model. The loss function for deep supervision is a weighted combination of losses at different scales, with 80% assigned to the loss of the final output and 20% assigned to the sum of losses from four intermediate outputs. In this approach, multiple intermediate $output_i$ (corresponding to different scales) are compared to their respective ground-truth labels $target_i$, in addition to the final $output$ which is also compared to the true $target$. This can be written as:

$$\begin{aligned} &output, output_1, output_2, output_3, output_4 \\ &= \text{MDFormer}(R, D) \\ total \\ &= 0.8 \times \text{loss}(output, target) \\ &+ 0.2 \sum_{i=1}^4 \text{loss}_i(output_i, F(target, output_i)) \end{aligned} \quad (6)$$

Where $F(x, y)$ is the interpolation function that maps x to the same spatial resolution as y and the specific form of the loss function is referred to E. Class Imbalance. In this setup, the intermediate auxiliary losses help guide the training by providing additional supervision at different scales.

E. Class Imbalance

The two datasets both share a common drawback: potential inhomogeneity in semantic annotation classes. To alleviate this issue, the following changes were made:

From the data perspective, we applied common augmentation techniques to expand the dataset. From the evaluation metrics perspective, in addition to OA [14], we introduced Macro F1 and Kappa [15], allowing for a more comprehensive and accurate assessment. From the loss function perspective, the model is ultimately trained using a combination of weighted cross-entropy loss, Dice loss, and Focal loss.

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \mathcal{L}_{Dice} + \mathcal{L}_{Focal} \quad (7)$$

III. EXPERIMENT

A. Experimental implements

Experiments were conducted using PyTorch on a single RTX 3080 GPU with 30GB of memory. The model is trained using the AdamW optimizer, with a learning rate of 0.001 for the decoder layers and 0.0005 for the encoder layers. A cosine annealing scheduler is applied, and training is conducted with a batch size of 32. To mitigate overfitting, weight decay is set to 0.0005. The channel split ratio in the CSFM is 4, while the kernel size and number in the SDEM are 3 and 2 convolution layers. The model was validated on Potsdam and Vaihingen. Patches are extracted with a window size of 256. In Potsdam, a stride of 64 pixels is used, while in the Vaihingen, a stride of 32 pixels is applied. The stride of the window determines the overlap between adjacent patches, allowing them to be directly input into the overlap patch embedding layer.

B. Performance Comparison

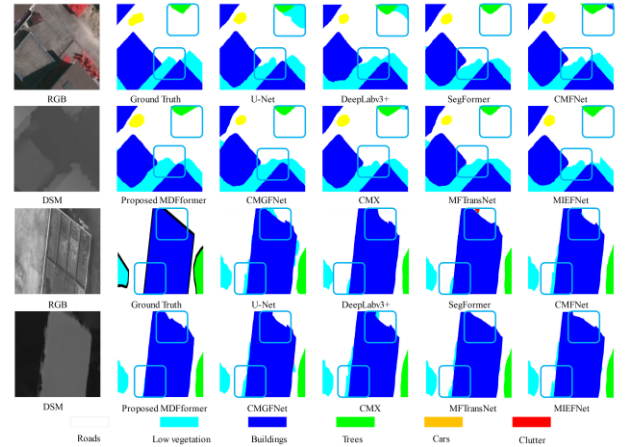


Fig. 2 Visualization results on Vaihingen(UP) and Potsdam(DOWN).

The results on the Potsdam dataset are shown in TABLE 1. As highlighted, MDFormer achieves the best OA of 91.25%, outperforming others in several categories, including Roads and Cars. In visual results Fig. 2, MDFormer demonstrates clear advantages in segmenting boundaries especially in comparison to other methods like DeepLabv3+ and CMX. This phenomenon can be attributed to the model's multi-stage extraction of multi-scale information, feature mixing in CSFM, DS to low-level features during training. Compared with models lacking emphasis on multi-scale integration, such as

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TABLE 1

COMPARATIVE RESULTS ON THE POTSDAM DATASET, BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS.

Method	CrossModal	OA(%)					Evaluation metrics			
		Roads	Buildings	Low veg.	Trees	Cars	M.F1	Kappa	MIoU	OA
U-Net [16]	N	90.73	96.27	87.12	82.96	92.58	85.39	86.89	77.45	88.74
DeepLabv3+[17]	N	90.29	96.33	87.08	82.3	91.06	84.5	86.06	76.57	87.42
SegFormer [18]	N	91.41	96.58	87.21	83.47	93.73	85.44	87.26	78.34	89.93
CMFNet [19]	Y	92.88	96.89	87.34	87.24	<u>96.26</u>	87.81	88.12	79.78	91.08
CMGFNet [4]	Y	92.75	<u>97.21</u>	87.83	85.36	96.11	87.48	88.04	79.62	90.85
CMX [12]	Y	<u>92.94</u>	97.06	<u>88.72</u>	85.17	95.94	87.35	87.92	79.48	90.72
MFTTransNet[19]	Y	92.21	96.80	87.93	84.26	95.37	87.65	88.11	<u>79.71</u>	90.92
MIEFNet [20]	Y	92.34	96.92	88.75	<u>87.35</u>	96.21	<u>87.85</u>	<u>88.23</u>	79.62	<u>91.1</u>
MDFformer	Y	93.28	97.42	87.01	87.47	96.57	87.96	88.41	79.98	91.25

TABLE 2

COMPARATIVE RESULTS ON THE VAIHINGEN DATASET, BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST RESULTS.

Method	CrossModal	OA(%)					Evaluation metrics			
		Roads	Buildings	Low veg.	Trees	Cars	M.F1	Kappa	MIoU	OA
U-Net [16]	N	92.27	94.1	77.52	90.14	78.47	86.5	87.02	78.79	90.11
DeepLabv3+ [17]	N	92.31	94.63	77.81	89.82	76.24	86.16	86.87	77.53	89.96
SegFormer [18]	N	92.74	95.82	78.92	90.36	82.66	88.1	87.72	79.46	90.61
CMFNet [19]	Y	92.9	96.94	80.86	90.32	85.32	89.26	88.48	81.16	91.43
CMGFNet [4]	Y	91.72	<u>97.21</u>	79.34	90.78	<u>88.16</u>	89.44	88.65	81.72	91.55
CMX [12]	Y	<u>93.04</u>	95.82	80.31	92.07	87.44	<u>89.75</u>	<u>89.07</u>	<u>82.53</u>	<u>91.72</u>
MFTTransNet [19]	Y	<u>92.50</u>	96.37	<u>80.92</u>	90.22	84.65	88.66	88.55	81.56	91.53
MIEFNet [20]	Y	92.96	97.05	80.56	90.11	85.56	89.71	89.05	82.48	91.69
MDFformer	Y	93.45	97.46	81.03	<u>91.07</u>	88.76	90.56	89.32	83.12	92.06

CMGFNet, MDFformer demonstrates superior capability in capturing both global contextual dependencies and local fine-grained details through its hierarchical feature interaction mechanism.

The final results on the Vaihingen are presented in TABLE 2. MDFformer achieves the best OA of 92.06%. Compared to single-modality methods, the use of RGB and DSM data gives MDFformer an advantage, especially in classes that benefit from elevation information, such as Low vegetation and buildings. The visual results Fig. 2 show a qualitative comparison across different methods. From these, it can be observed that MDFformer consistently provides better segmentation performance, especially around boundaries.

C. Model Complexity Analysis

We conducted a comprehensive complexity analysis of MDFformer (TABLE 3), benchmarking it against representative methods in the field. The results demonstrate that MDFformer achieves a favorable balance between computational efficiency and segmentation accuracy. MDFformer achieves 47.33G FLOPs, striking a balance between U-Net and CMFNet, and has 35.82M parameters to outperform U-Net (62.85M) and V1 (61.51M). These advantages stem from its lightweight design (leveraging weight connections, channel shuffle, and deep supervision) and efficient Adaptive Frequency Kernel for linear complexity, which together enable effective multi-scale feature extraction without excessive resource consumption.

D. Ablation Study

TABLE 4 presents the results of the ablation study, highlighting the contributions of key components. When CSFM is

TABLE 3

COMPUTATIONAL COMPLEXITY MEASURED BY A 256×256 IMAGE. MIOU VALUES WERE THE RESULTS ON THE VAIHINGEN DATASET. V1 AND V2 DENOTE THE VARIANTS OF MDF-FORMER AFTER REPLACING THE BACKBONE.

Method	Flops (G)	Parms (M)	Fps	MIoU
U-Net [16]	22.96	62.85	54.08	78.79
DeeplabV3+ [17]	16.54	59.34	<u>42.59</u>	77.53
Segformer [18]	3.02	13.72	29.62	79.46
CMFNet [19]	78.25	98.49	11.89	81.16
CMGFNet [4]	83.05	170.24	28.62	81.72
CMX [12]	10.58	65.55	18.3	82.53
MFTTransUNet[19]	<u>8.44</u>	43.77	14.88	81.56
MIEFNet [20]	19.42	56.21	15.38	82.48
V1 [18]	11.57	61.51	12.46	<u>83.1</u>
V2 [21]	20.23	40.01	17.73	82.3
MDFformer	47.33	<u>35.82</u>	17.24	83.12

included, the performance shows a clear improvement in M.F1 and Kappa scores, demonstrating the importance of cross-modal feature fusion. Removing the SDEM results in a noticeable drop, emphasizing the need for spatial detail enhancement in multi-modal data. Lastly, the absence of DS leads to a decrease in performance, indicating that additional supervision at intermediate layers is crucial for optimizing feature learning. These results confirm that each module in MDFformer contributes uniquely, enhancing the model’s overall performance.

IV. CONCLUSION AND FUTURE WORK

In this work, MDFformer is proposed, which effectively performs long-range modeling between two modalities and different scales, making it suitable for remote sensing segmentation

TABLE 4
ABLATION STUDY ON TWO DATASETS

Modules			Datasets							
			Vaihingen				Potsdam			
CSFM	SDEM	DS	M.F1	Kappa	MIoU	OA	M.F1	Kappa	MIoU	OA
✓	✓	✓	90.56	89.32	83.12	92.06	87.96	88.41	79.98	91.25
✓	✓		90.41	89.21	82.96	91.95	87.95	88.35	79.89	91.19
✓		✓	89.84	89.06	82.65	91.79	87.87	88.12	79.72	91.09
	✓	✓	90.19	89.15	82.76	91.89	87.92	88.19	79.86	91.13
✓			89.49	88.82	82.22	91.65	87.59	88.15	79.51	90.89
	✓		89.76	88.99	82.39	91.72	87.48	88.01	79.56	90.92
		✓	89.47	88.68	81.72	91.56	87.37	87.92	79.42	90.75

tasks. And strong performance is demonstrated on Vaihingen and Potsdam. MDFformer utilizes innovative modules CSFM and SDEM to enhance feature fusion and spatial detail, along with DS. SpectrumRecombineFormer [22] and Data augmentation [23] provide methodologies for combining diverse data sources, which can enhance the performance. Furthermore, PrefGNN [24] and hypergraph [25] offer advanced techniques for modeling complex relationships and incorporating additional information. These could be adapted to improve the accuracy in future directions.

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